

MYS KAMENNYJ, Russian Federation

WMO: 231460

Lat: **68.500N**Lon: **73.583E**Elev: **5**StdP: **101.26**Time Zone: **5.00 (E05)**Period: **86-94**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.6	-39.5	-43.8	0.1	-41.1	-41.8	0.1	-39.2	18.8	-4.0	16.5	-17.3	3.4	60	1.086

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	19.1	16.3	17.0	14.4	15.1	13.0	16.7	18.7	14.6	16.4	13.3	14.8	6.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.2	18.4	13.8	9.8	15.8	12.4	8.9	14.5	47.0	18.8	41.3	16.6	37.5	14.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.4	12.9	DB	-43.8	21.7	3.9	3.7	-46.6	24.4	-48.9	26.5	-51.1	28.6	-53.9	31.3	(4)
(5)			WB	-43.2	17.9	2.9	3.2	-45.3	20.1	-47.0	22.0	-48.7	23.8	-50.8	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.7	-25.9	-25.6	-17.8	-15.6	-4.5	2.3	10.7	9.9	4.6	-5.6	-15.6	-22.7
	DBStd	14.55	8.16	9.39	8.61	8.08	4.89	3.40	4.18	3.12	3.32	6.25	8.33	8.55
	HDD10.0	6932	1112	996	862	768	451	232	40	42	164	484	769	1012
	HDD18.3	9876	1371	1229	1120	1018	709	481	240	263	413	742	1019	1271
	CDD10.0	102	0	0	0	0	0	1	62	38	1	0	0	0
	CDD18.3	4	0	0	0	0	0	0	4	0	0	0	0	0
	CDH23.3	17	0	0	0	0	0	0	17	0	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	6.3	6.4	6.0	6.6	6.5	6.1	5.4	5.5	6.1	6.8	7.5	6.5	6.5
Precipitation	PrecAvg	389	24	22	20	22	28	29	36	55	50	32	26	25
	PrecMax	613	54	87	57	65	98	80	106	142	96	77	58	76
	PrecMin	233	3	1	2	1	10	1	2	5	19	13	8	3
	PrecStd	83	12	17	13	14	16	17	27	30	19	13	11	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.7	-4.5	-0.6	1.7	3.4	14.2	25.6	18.4	12.0	7.2	0.9	-2.9
		MCWB	-6.3	-5.1	-1.2	0.7	2.0	11.4	21.0	15.6	10.1	6.1	0.2	-3.5
	2%	DB	-9.2	-7.0	-1.9	0.9	2.4	10.1	21.4	16.5	11.0	4.8	-0.8	-4.8
		MCWB	-9.6	-7.5	-2.4	0.2	1.3	8.5	18.8	14.4	9.6	4.1	-1.3	-5.2
	5%	DB	-11.6	-8.5	-3.1	0.3	1.6	8.1	18.7	15.3	10.2	3.2	-2.0	-7.5
		MCWB	-12.0	-9.0	-3.5	-0.3	0.7	6.5	16.1	13.3	9.0	2.7	-2.4	-8.0
	10%	DB	-14.0	-11.2	-4.8	-1.6	1.1	6.5	17.0	14.4	9.2	1.5	-3.4	-9.6
		MCWB	-14.4	-11.6	-5.3	-2.1	0.3	5.0	14.4	12.6	8.1	1.0	-3.8	-10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.9	-5.0	-1.1	1.1	2.5	11.6	21.8	16.2	10.6	6.2	0.3	-3.3
		MCDB	-5.6	-4.6	-0.8	1.4	3.0	13.3	25.5	17.8	11.5	7.0	0.7	-2.9
	2%	WB	-9.6	-7.4	-2.3	0.5	1.6	8.6	18.8	14.6	9.9	4.1	-1.2	-5.1
		MCDB	-9.3	-7.1	-2.0	0.9	2.2	10.1	21.3	16.0	10.7	4.8	-0.9	-4.7
	5%	WB	-11.9	-9.0	-3.5	-0.3	1.0	6.6	16.4	13.7	9.2	2.6	-2.4	-7.9
		MCDB	-11.6	-8.5	-3.2	0.3	1.6	7.9	18.4	15.0	10.1	3.1	-2.0	-7.5
	10%	WB	-14.4	-11.7	-5.2	-2.1	0.4	5.2	14.6	12.9	8.2	1.1	-3.9	-9.9
		MCDB	-14.1	-11.4	-4.8	-1.6	1.1	6.3	16.5	14.0	9.2	1.4	-3.5	-9.5
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.6	8.7	9.6	5.4	4.2	5.4	3.9	3.1	4.8	6.8	8.0
		MCDBR	12.2	9.0	10.4	7.7	3.8	7.4	7.4	5.1	3.8	3.8	7.0	8.8
		MCWBR	12.0	9.1	9.8	7.5	3.5	6.0	5.5	4.0	3.2	3.5	6.7	8.6
	5% WB	MCDBR	12.4	9.5	9.5	7.5	3.7	7.2	7.0	4.6	3.5	3.8	6.9	8.5
		MCWBR	12.1	9.4	9.5	7.4	3.6	6.1	5.8	3.9	3.3	3.5	7.0	8.3
Clear-Sky Solar Irradiance	taub	0.210	0.252	0.302	0.324	0.342	0.360	0.386	0.370	0.319	0.265	0.205	N/A	
	taud	1.865	1.988	2.029	2.066	2.134	2.314	2.344	2.371	2.450	2.200	1.936	N/A	
	Ebn at Noon	50	499	697	806	838	836	796	764	718	502	49	0	
	Edh at Noon	8	50	90	117	123	107	102	88	62	40	6	0	
All-Sky Solar Radiation	RadAvg	0.02	0.40	2.00	4.07	5.74	5.54	5.00	3.16	1.61	0.53	0.08	0.00	
	RadStd	0.00	0.06	0.17	0.32	0.37	0.29	0.62	0.42	0.15	0.07	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page